

Technical Document: Rail Exit Velocity

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IREC 25-26 Propulsion

Purpose/Objective

For the IREC 25-25 30K SRAD competition the propulsion team made the decision to change the motor tube casing from 4" diameter to a 6" diameter. Consequently, the motor increases greatly in mass when loaded with solid propellant. This increase in mass makes it more difficult to reach the required speed off the launch rod. Reaching this goal becomes much more difficult. To help solve this issue the following script was developed to calculate the maximum allowable mass for a rocket to give a specific designed engine. This script takes in two motor files and graphs them against each other to determine weight vs velocity curves for both.

How the Script Works

The script was developed to calculate the desired rocket mass given a motor thrust curve and rail velocity requirement. This problem is complicated as it cannot be solved arithmetically, instead numerical methods need to be used to help calculate the mass. The reason arithmetic cannot be used to find the mass is due to the dependence of both velocity and the time on the rod is to mass. When trying to use a simple impulse equation or kinematics, an assumption is needed that greatly alters the amount of mass allowed on the rocket. Because of this, the script uses an ordinary differential equation to help solve for maximum allowable mass.

The script works fundamentally based on the equation outlined below:

$$a = \frac{d^2x}{dt^2} = (T(t) - mg)/m$$

Where the acceleration is equated to the thrust w.r.t. time (T(t)) minus the force of gravity on the vehicle divided by the mass. The mass of the vehicle is assumed constant, therefore this equation underestimates the acceleration of the vehicle and thus the required minimum mass

slightly. This ordinary differential equation helps the script solve for what velocity would be off the rod based off of the initial conditions:

$$x(0) = 0, \quad \frac{dx}{dt} @ (t = 0) = 0$$

Once the above equations are defined, MATLAB can be used to create the following code as a function of position. Once this is done, the *ode45()* function can be used to solve the system. To use this function an event must first be defined to stop analysis once the rocket leaves the rod. In the code this is known as the *railExit()* function and is saved as an option applied to the *ode45* command.

It is important to note that this methodology of calculating the mass does not account for the angle of the rod. While this may seem alarming the disregarding the angle of the launch provides a more conservative estimate for the mass of the rocket. When the rocket is angled, gravity is not able to work against the rocket as greatly. Because of this, the script has a more conservative estimate for the mass of the rocket.

Now that the computational methodology is understood the user will be able to run the code with MATLAB and select any properly formatted .eng file to determine the amount of mass the rocket must be to reach the set velocity at the end of the rod.

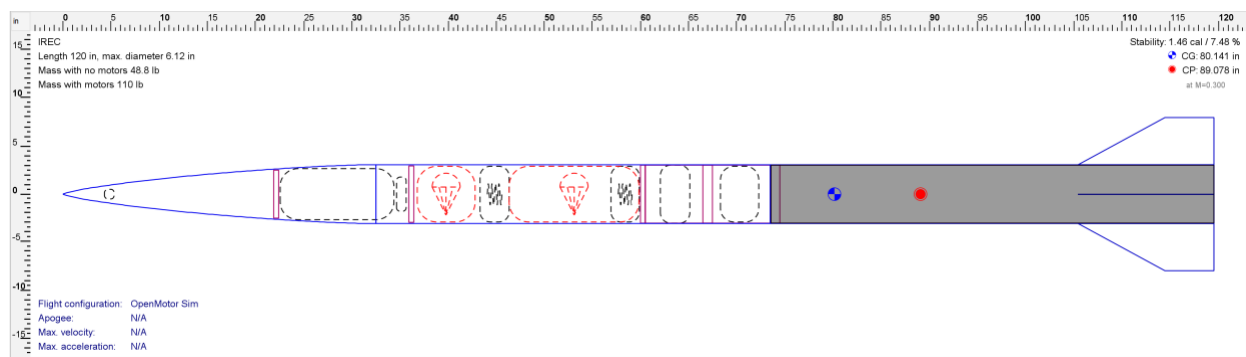
Verification

Below is proof of the functionality of the script. To verify this functionality the scripts output is compared to that of an OpenRocket simulation.

When using the **MRB_11.12.25_OM.eng** file and setting the launch speed off the rod at 25 m/s the script outputs the following:

“The maximum weight allowed for the O4727 motor is 110.022892 lbm”

This reveals that if we create a rocket with the O4727 motor provided in the file, the launch speed off the rod will be roughly be equivalent to 25 m/s. To verify this modified version of the 25/26 IREC competition rocket was created in OpenRocket with recommended weight.



As shown above the weight of the rocket is roughly 110lb with propellant. The velocity curve from this rocket and selected motor was then analyzed with an effective launch rail length of 13 ft and displayed the following:

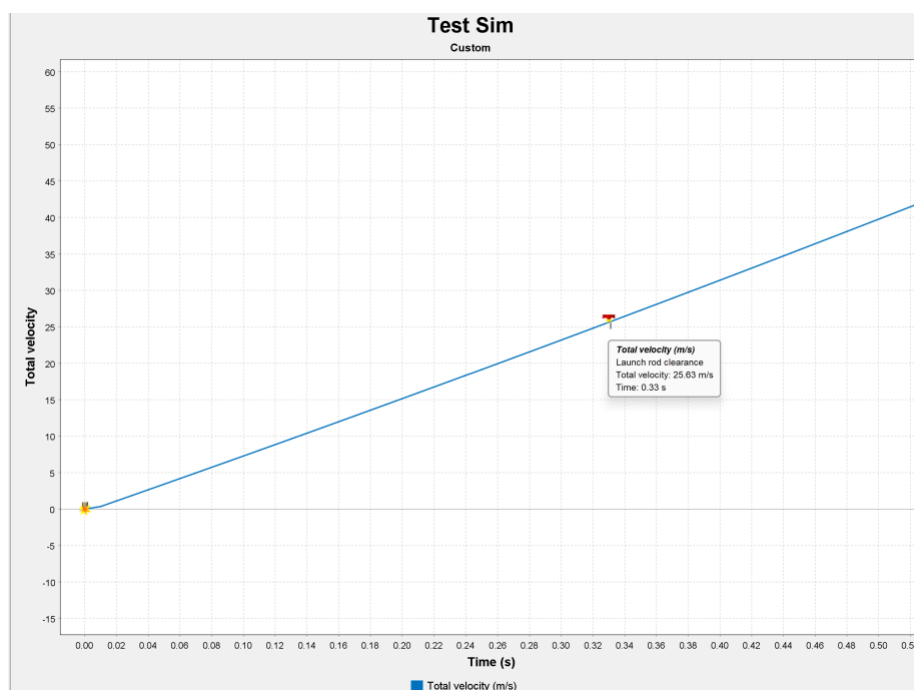


Figure 1: Rocket Velocity vs Time

Since the recommended weight almost matched the set velocity off the rod with a slight underprediction the script has been verified.

Conclusion

This script will be useful as tool when developing future grain configurations. Users will no longer need to calculate these numbers by hand and instead can use the script to verify the speed of the rod based.